Aravindh Varadharaj

Go Backend Engineer · Concurrency · Payments · Distributed Systems · Data Pipelines

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Live resume: aravindh97.github.io/resume (ask my code questions, kill the Raft leader, flip sync.Mutex on a data race). Every project below has a live demo: its Go code compiled to WebAssembly, running in the browser.

SUMMARY

Builder at Amazon since 2021 who moved from digital marketing into engineering by automating my team's workflows. I build Go backend systems and prove they hold up: concurrency under load, exactly-once money movement, crash and outage tests, offline sync, streaming pipelines and performance work, each with tests, CI and measured results against a baseline.

SKILLS

Go: goroutines, channels, context, sync, net/http, generics, WebAssembly, race detector, benchmarks and profiling
Data: PostgreSQL (transactions, row locking, SKIP LOCKED, advisory locks, partial unique indexes, logical replication, pgvector), OpenSearch, Redis basics
Reliability: idempotency keys, retries with backoff, dead-letter states, Temporal workflows, kill -9 and outage testing, watermarks and backpressure
Distributed systems: Raft, CRDTs, version vectors, hybrid logical clocks, CDC
Quality: evals and backtests with baselines, chance and held-out data, integration tests on Docker
AI coding tools: Kiro, Claude Code and Copilot daily; concurrency, SQL and retry logic reviewed by hand

PROJECTS

fairdrop: Ticketing for High-Demand Releases

Live demo · GitHub

Go, PostgreSQL, drand public randomness

- Seat holds claim exactly the requested seats or none in one locked UPDATE (fixed lock order, SKIP LOCKED); payments are idempotent by key; families are seated together.
- 50,000 simulated buyers with timeouts, retries, double clicks and abandoned holds: 0 seats sold twice in 3 runs vs 156,874 extra bookings for the naive version; charges always equal bookings plus refunds.
- Queue order comes from a future drand round committed before registration closes, so any buyer can verify it; swapped positions and late entries are detected.

paisa: Exactly-Once Double-Entry Ledger

Live demo · GitHub

Go, PostgreSQL

- Idempotency key, postings and balances written in one transaction; retries return the stored outcome, including rejections; reusing a key for a different request is refused.
- kill -9 the payment process 20 times during 50,000 payments: books balanced after every kill and final balances of all accounts matched a clean run exactly.
- Reconciliation found 477 of 477 planted bank-file mismatches and fixed them idempotently; sharding the hot fee account raised throughput 6.8x (437 to 2,981 payments/s).

Granth: Document Ingestion & Hybrid Retrieval for Agents

Live demo · GitHub

Go, Temporal, OpenSearch, PostgreSQL, Tesseract OCR, MCP

- One Temporal workflow per document (fetch, parse, OCR fallback, chunk, embed, index). Activities pass sha256 blob references and chunk ids are deterministic, so retries and replays never create duplicates.
- Corrupt, unsupported or oversized files fail once and go to a Postgres dead-letter ledger with the failing step. Transient failures retry with backoff for up to an hour; an embedding cache stops retries from paying twice.
- Stopped OpenSearch mid-ingest in a live test: the index step retried 7+ times and the document indexed after restart with no duplicate chunks.

Smriti: Long-Term Memory for LLM Agents

Live demo · GitHub

Go, PostgreSQL + pgvector, Temporal, MCP

- User, team and org memories with one access rule applied in every query. Org writes need explicit permission, so one user's chat cannot change what every agent believes.
- Each fact is added, confirmed or superseded instead of appended. An advisory lock plus a partial unique index keep one current value per subject (tested with 20 concurrent writers); replaying a conversation is a no-op.
- Bitemporal rows keep history and source message, so agents can answer both "what is true now" and "what did we believe in February".

Sutra: Enterprise Connectors, Ontology & Entity Resolution

Live demo · GitHub

Go, PostgreSQL logical replication (pgoutput), REST, YAML ontology, MCP

- CDC from Postgres with an exported-snapshot initial load and commit-time checkpoints. After a kill -9 mid-stream and restart, source and warehouse matched exactly (md5 over every row).
- Incremental CRM sync with keyset pagination, Retry-After handling and checkpoints written in the same transaction as each page.
- Entity resolution with GSTIN checksum validation, PAN derived from GSTIN, +91 phone normalisation and transliteration-aware names (Sri/Shree, Lakshmi/Laxmi). Two different PANs can never merge; unsure pairs go to review.

orbit: Satellite Conjunction Screener Live demo · GitHub

Go, SGP4, CelesTrak public catalogue

- Altitude shells, a spatial hash sized from the maximum closing speed, and a straight-line prefilter before golden-section refinement: checks 0.01% of pairs, found 190 of 190 encounters an exhaustive brute force found.
- 16,763 objects over 6 days in 3 min 49 s on 8 cores (3.8 M object-steps/s); fixed an epoch bug in the upstream SGP4 port (median 3.6 km error), now within 1 mm of reference.

relief: Offline-First Sync for Disaster Relief Teams Live demo · GitHub

Go, CRDTs, hybrid logical clocks

- Counters for stock, add-wins needs and HLC last-writer-wins fields over an op log; version vectors send 3.3% of the data a full copy would; works over USB bundles.
- 10,000 randomised runs with partitions, relays, duplicates and clock skew: every device converged with 0 rule violations; last-writer-wins sync lost data in all 10,000.

aqcal: Low-Cost Air Sensor Calibration and Fault Detection Live demo · GitHub

Go streaming pipeline, real Berlin sensor and UBA reference data

- Sharded aggregation with watermarks and bounded channels: 6.9 M readings/s on 8 shards.
- Per-sensor calibration cut hourly PM2.5 error 52% (4.81 to 2.31 ug/m3) on held-out days; detectors caught 298 of 300 injected faults.

foodwatch: Food-Stress Early Warning (Somalia) Live demo · GitHub

Go, CHIRPS rainfall, WFP prices, IPC classifications

- Backtested against IPC crisis classifications with no lookahead: the rainfall signal flagged 20 of 26 regional crisis onsets (10.7 expected by chance) a median 21.6 weeks early.
- Cross-checked rainfall sources: NASA POWER called the failed 2021 Deyr season wet, so the backtest uses CHIRPS.

Game: Pallanguzhi (Tamil mancala) against a Go alpha-beta search AI that explains its moves, playable in the browser; a reproducible 2,800-game parallel benchmark showed the second player has a large advantage, so results are reported by seat.

Earlier projects: In github.com/Aravindh97/aravindh-impact-portfolio: OpenSchool Enterprise (Python/FastAPI learning platform with role contracts and auditable review state), FamineLens (sourced historical evidence with an uncertainty-aware scenario model), Kindred (privacy-aware animal welfare network with a human-review queue), EcoSignal (Go RF power-plan optimiser under a coverage constraint), Nuclear Legacy & Civil Safety (Go observatory).

EXPERIENCE

Sr. Digital Marketing Associate, Amazon 2021 to Present

- Built and automated internal tools and workflows for my team, replacing repetitive manual tasks with reliable, reusable solutions.
- Worked with stakeholders to find process gaps, then designed, built and supported automation that the team adopted in daily work.
- Grew from a marketing role into hands-on development, and built a portfolio of Go and Python backend systems outside my core responsibilities.

EDUCATION

Bachelor of Engineering (B.E.) in Mechanical Engineering